

NITHIN VASAN G A

Mechatronics Engineer | AI & Automation Enthusiast

📍 Coimbatore, Tamil Nadu, India | 📞 +91 99408 19979 | ✉ nithinvasan2007@gmail.com

[LinkedIn: linkedin.com/in/nithin-vasan-23918532a](https://www.linkedin.com/in/nithin-vasan-23918532a)

PROFESSIONAL SUMMARY

A highly motivated 2nd-year B.E. Mechatronics Engineering student at Kumaraguru College of Technology with a strong foundation in automation, artificial intelligence, robotics, and mechanical design. Proficient in industry-standard CAD tools including AutoCAD and SolidWorks, complemented by hands-on exposure through internships in generative AI integration within mechatronics domains. Passionate about bridging the gap between intelligent software and smart hardware to drive the future of Industry 4.0. Seeking opportunities as an Automation Engineer, AI-Integrated Robotics Developer, or CAD/CAM Design Engineer to contribute to innovative engineering solutions.

EDUCATION

B.E. Mechatronics Engineering

Kumaraguru College of Technology, Coimbatore

2024 – 2028 | CGPA: 8.23 / 10

Relevant Coursework: Robotics & Automation, Control Systems, Mechatronics Design, Machine Learning Applications, PLC & SCADA, CAD/CAM

Higher Secondary Education (11th & 12th Grade)

Lakshmi Matric Hr. Sec. School, Manaparai, Trichy

Completed: 2024

Stream: Physics, Chemistry, Mathematics, Computer Science

INTERNSHIP EXPERIENCE

Engineering Intern

Robotica FA Solutions

210 Hours | Industrial Automation Domain

- ▶ Gained hands-on exposure to industrial robotic systems, automation hardware, and factory automation (FA) solutions.
- ▶ Studied PLC-based control systems and real-time process automation workflows in manufacturing environments.
- ▶ Explored integration of AI/ML modules for predictive maintenance and process monitoring in automation setups.
- ▶ Collaborated with senior engineers on system design documentation and troubleshooting of automated machinery.

AI & Technology Intern

Bluecode Academy

210 Hours | AI & Software Domain

- ▶ Acquired core competencies in Generative AI concepts including large language models, prompt engineering, and AI pipelines.
- ▶ Developed prototypes integrating GenAI tools (ChatGPT API, Gemini) into mechatronics workflow automation.
- ▶ Designed AI-assisted diagnostic systems for smart fault detection in mechanical and electronic sub-systems.
- ▶ Completed project on embedding AI-driven decision making into CNC machine process control parameters.

PROJECTS

AI-Integrated Process Parameter Detection in CNC Machines

Technologies: *Python, TensorFlow/Scikit-learn, OpenCV, MATLAB, CAD/CAM Software*

Developed an AI-powered system to automatically detect and optimize key CNC machining parameters—including cutting speed, feed rate, depth of cut, and tool path—to maximize surface finish quality, material removal rate (MRR), and overall machining efficiency. The system uses sensor data and machine learning models trained on historical machining datasets.

Achievement: Achieved ~18% improvement in surface roughness (Ra) prediction accuracy and a 12% reduction in tool wear through optimized parameter recommendations, demonstrating measurable efficiency gains over manual tuning.

Autonomous Line-Following Robot with AI Obstacle Avoidance

Technologies: *Arduino, Raspberry Pi, Python, OpenCV, Ultrasonic/IR Sensors*

Designed and built a mechatronics-based autonomous robot capable of following predefined paths using IR sensor arrays and avoiding dynamic obstacles in real time using computer vision and ultrasonic ranging. Integrated PID control loops for smooth navigation and an OLED display for live telemetry.

Achievement: Successfully navigated complex track configurations with 95% path accuracy; project recognized as a Best Innovation entry in the college-level technical symposium.

Smart Home Automation System using IoT and Voice AI

Technologies: *ESP32, Python, MQTT, Google Assistant API, Node-RED, SolidWorks (enclosure design)*

Engineered a cost-effective IoT-based home automation system enabling voice-controlled and mobile app-controlled management of electrical appliances. Designed the custom housing enclosure in SolidWorks and integrated the electronic module for seamless deployment. Used MQTT protocol for low-latency device communication.

Achievement: Reduced manual intervention by 70% in a prototype household setup; designed and 3D-printed the enclosure achieving a near-production quality finish with SolidWorks-generated models.

SKILLS

Programming & AI: Python • C++ • MATLAB • Machine Learning (Scikit-learn, TensorFlow) • Prompt Engineering & GenAI Integration

CAD / Design: SolidWorks • AutoCAD • 3D Modeling & Simulation • CAD/CAM Workflows

Automation & Robotics: PLC Programming (Ladder Logic) • Arduino & Raspberry Pi • Sensor Integration • IoT Systems

Tools & Software: MATLAB/Simulink • Git & GitHub • Node-RED • VS Code • MS Office / Excel

Soft Skills: Analytical Problem-Solving • Team Collaboration • Technical Communication • Adaptability • Leadership

ACHIEVEMENTS

- ▶ Best Innovation Award – Secured recognition for the Line-Following Robot with AI Obstacle Avoidance at KCT’s inter-department technical symposium, competing against 30+ student teams.
- ▶ Top 10 Finalist – Selected among the Top 10 teams in a state-level Smart India Hackathon (SIH) internal round for the AI-CNC integration project proposal, out of over 80 participating teams.
- ▶ Academic Excellence – Maintained a CGPA of 8.23 / 10 in the first year of B.E. Mechatronics Engineering, placing in the top 15% of the batch at Kumaraguru College of Technology.

LANGUAGES

Tamil (Native) **English** (Professional Proficiency) **Japanese** (Elementary / Conversational)

INTERESTS & HOBBIES

 **Chess** (Strategic thinking & competitive play)  **Gaming** (Simulation & strategy genres)  **Yoga** (Mindfulness & physical wellness)